

//SIT55 : OOP 1 : COMPLEX NO.OPERATIONS

```
package Javaprogs;
import java.util.Scanner; /*Scanner is used to receive user input and parse(to read the value of one
object to convert it to another type)
them into data types such as int, double or default String */
class Complex
{
    float real,img; //data members
    public Complex() /*default constructor to assign value to variables*/
    {
        real=0;
        img=0;
    }
    public Complex(float a ,float b) //parameterized constructor accepts arguments or parameters
    {
        real=a;
        img=b;
    }
    public void Display(Complex C1,Complex C2) //function to display complex no entered by user
    {
        System.out.println("FIRST COMPLEX NUMBER = "+C1.real+" "+C1.img+"i");

        System.out.println("SECOND COMPLEX NUMBER = "+C2.real+" "+C2.img+"i");

    }

    public void Add(Complex C1,Complex C2)
    {
        float real,img;
        real=(C1.real+C2.real);
        img=(C1.img+C2.img);
        System.out.println("ADDITION OF TWO COMPLEX NUMBERS IS = "+real+" "+img+"i");
    }
    public void Sub(Complex C1,Complex C2)
    {
        float real,img;
        real=(C1.real-C2.real);
        img=(C1.img-C2.img);
        System.out.println("SUBTRACTION OF TWO COMPLEX NUMBERS IS =
"+real+" "+img+"i");
    }
    public void Mult(Complex C1,Complex C2)
    {
        float real, img;
        real = (C1.real * C2.real - C1.img * C2.img);
        img = (C1.real * C2.img + C1.img * C2.real);
        System.out.println("MULTIPLICATION OF TWO COMPLEX NUMBERS IS =
"+real+" "+img+"i");
    }
}
```

```

    }
    public void Div(Complex C1,Complex C2)
    {
        float real,img;
        real=(C1.real*C2.real+C1.img*C2.img)/(C2.real*C2.real+C2.img*C2.img);
        img=(C1.img*C2.real-C1.real*C2.img)/(C2.real*C2.real+C2.img*C2.img);
        System.out.println("DIVISION OF TWO COMPLEX NUMBERS IS = "+real+"+"+img+"i");
    }

}

public class Main
{
    public static void main(String[] args)
    {

        float num1,num2;

        Scanner sc=new Scanner(System.in);
        Complex cal=new Complex();
        System.out.print("ENTER REAL PART OF FIRST NUMBER : ");
        num1=sc.nextFloat(); //This method reads a float value input scanned from the user
        System.out.print("ENTER IMAGINARY PART OF FIRST NUMBER : ");
        num2=sc.nextFloat();
        Complex C1=new Complex(num1,num2);

        System.out.print("ENTER REAL PART OF SECOND NUMBER : ");
        num1=sc.nextFloat();
        System.out.print("ENTER IMAGINARY PART OF SECOND NUMBER : ");
        num2=sc.nextFloat();
        Complex C2=new Complex(num1,num2);

        System.out.println();
        cal.Display(C1,C2); //calling display function

        System.out.println();
        cal.Add(C1,C2); //calling addition function

        cal.Sub(C1,C2); //calling subtraction function

        cal.Mult(C1,C2); //calling multiplication function

        cal.Div(C1, C2); //calling division function

    }

}

```

/*OUTPUT :

ENTER REAL PART OF FIRST NUMBER : 6

ENTER IMAGINARY PART OF FIRST NUMBER : 3

ENTER REAL PART OF SECOND NUMBER : 4

ENTER IMAGINARY PART OF SECOND NUMBER : 2

FIRST COMPLEX NUMBER = $6.0+3.0i$

SECOND COMPLEX NUMBER = $4.0+2.0i$

ADDITION OF TWO COMPLEX NUMBERS IS = $10.0+5.0i$

SUBTRACTION OF TWO COMPLEX NUMBERS IS = $2.0+1.0i$

MULTIPLICATION OF TWO COMPLEX NUMBERS IS = $18.0+24.0i$

DIVISION OF TWO COMPLEX NUMBERS IS = $1.5+0.0i$ */